

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

Project portfolio budget allocation for EPC contractors in the oil and gas sector is a sequential decision-making problem under uncertain cash inflows, cash outflows, and project performance. Each portfolio consists of multiple projects with heterogeneous attributes, including budget at completion, start and finish dates, contractual S-curves, fixed profit margins, milestone-based payment structures, and uncertain contractor performance represented through Earned Value Management metrics such as cumulative SPI and CPI. The model also captures delayed and uncertain milestone payments, working-capital shortages, credit usage during negative-cashflow periods, and management intervention actions that affect cashflow and performance trajectories.

Despite its practical importance, rigorous mathematical modeling of project portfolio budgeting under joint cashflow and performance uncertainty remains largely absent in recent literature, while most AI/ML-related studies discuss the topic qualitatively. This paper proposes a custom stochastic reinforcement learning environment for learning adaptive budget allocation policies under unknown future uncertainty. A PPO-based agent is trained with a multi-objective reward function to allocate constrained budgets efficiently, prioritize economically viable projects, avoid loss-making allocations, and learn optimal allocation timing. The framework provides a foundational RL model for EPC oil and gas portfolio budgeting, supports future fine-tuning with anonymized industry data, and enables immediate next-period allocation recommendations as well as rolling 12-period trajectory simulation.

Keywords: Project Portfolio Budgeting, Reinforcement Learning, EPC Contractors, Oil and Gas Projects, Cashflow Uncertainty, Earned Value Management, PPO, Rolling Horizon Control

1 Introduction

1.1 Motivation and Context

Project portfolio management (PPM) is a fundamental organizational capability for maintaining competitive advantage and achieving strategic objectives. In project-based industries, portfolio decisions determine which initiatives are selected, prioritized, and funded

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under limited resources, conflicting criteria, and uncertain execution outcomes (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Martinsuo, 2013; Killen et al., 2008). These decisions become particularly critical in large-scale engineering environments, where organizations must coordinate multiple concurrent projects while balancing strategic value, financial feasibility, schedule pressure, and risk exposure.

In the Engineering, Procurement, and Construction (EPC) sector, especially in oil and gas projects, portfolio budgeting is complicated by heterogeneous contract structures, long project durations, capital intensity, and uncertain cash-flow behavior (Merrow, 2011; Cui et al., 2010; Kenley and Wilson, 1986; Kaka and Price, 1993). Budget allocation must account for both cash outflows associated with procurement, labor, equipment, and sub-contractors, and delayed cash inflows tied to contractual milestones and client approvals. Consequently, portfolio budgeting is not merely a one-time allocation problem; it is a sequential decision process in which current funding choices influence future project progress, liquidity conditions, and portfolio-level performance.

This challenge is amplified by persistent uncertainty in project execution. Large industrial and infrastructure projects frequently experience delays and cost overruns, and such deviations can propagate through both schedules and financial plans (Flyvbjerg et al., 2002, 2003; Love et al., 2013, 2016). Earned value indicators such as the Schedule Performance Index (SPI) and Cost Performance Index (CPI) provide a compact representation of execution performance and can support project monitoring and control (Fleming and Koppelman, 2016; Lipke et al., 2009; Vanhoucke, 2012). However, these indicators also underscore the difficulty of making reliable budget allocation decisions when future project states, payment realizations, and performance deviations remain uncertain. Therefore, portfolio budgeting under uncertain cash flows requires methods that can reason across multiple projects, multiple time periods, and stochastic state transitions.

The broader literature on project portfolio management similarly characterizes such environments by large data volumes, multiple decision criteria, resource constraints, and the need for investment optimization. These characteristics are consistent with prior research on project portfolio selection, project scheduling under uncertainty, and construction cash-flow forecasting (Gutjahr and Froeschl, 2013; Doerner et al., 2006; Khanzadi et al., 2018; Golenko-Ginzburg and Gonik, 1997; Ballestín and Blanco, 2015; Barraza et al., 2007). Taken together, they motivate the development of intelligent decision-support methods when static or manually driven budgeting approaches become insufficient.

1.2 Research Gap

A substantial body of literature has addressed project portfolio selection, prioritization, and scheduling through operations research, multi-criteria decision-making, stochastic programming, and metaheuristic optimization (Archer and Ghasemzadeh, 1999; Liesiö et al., 2007; Klerides and Hadjiconstantinou, 2019; Elshaer, 2014). These methods provide valuable structure for complex decision problems, but they are often less suitable when portfolio budget allocation must respond to evolving project conditions under uncertainty. In many existing formulations, project cash flows, budget requirements, and performance trajectories are treated as fixed inputs, deterministic plans, or exogenous scenarios. This limits their ability to capture the interaction between current allocation decisions and future project and financial states.

A central limitation in this context is that portfolio budgeting in EPC environments is inherently sequential. Budget releases made at one time step affect project execution, milestone realization, cash availability, and subsequent allocation opportunities. Yet much of the existing literature approaches project portfolio decisions as static optimization,

stage-wise planning, or scheduling problems rather than as a learned sequential policy problem under uncertainty. As a result, the adaptive linkage between observed portfolio state and subsequent allocation action remains insufficiently modeled.

Recent advances in machine learning and reinforcement learning provide a promising direction for this class of problems. Reinforcement learning is specifically designed for sequential decision-making under uncertainty and has demonstrated value in operations, scheduling, and process control settings (Puterman, 1994; Sutton and Barto, 2018; Wauters et al., 2016; Chen et al., 2018; Hubbs et al., 2020; Bengio et al., 2021). Nevertheless, its application to project portfolio budgeting under uncertain cash flows remains limited. There is still a need for a reproducible framework that can integrate project-level uncertainty, cash-flow timing, milestone-based payments, and adaptive resource allocation within a portfolio-level sequential decision architecture.

A further practical constraint is data availability. Detailed industrial project portfolio data are typically confidential and difficult to release publicly, which creates a barrier to reproducible experimentation. As a result, the literature lacks sufficiently transparent benchmark environments for testing adaptive portfolio budgeting policies. This gap motivates the use of literature-calibrated synthetic data as a practical foundation for developing and evaluating learning-based decision-support systems (Jordon et al., 2022; Xu et al., 2019).

1.3 Proposed Approach and Contributions

To address these gaps, this paper develops a deep reinforcement learning framework for optimal project portfolio budgeting under uncertain cash flows. The problem is modeled as a sequential decision process in which an agent observes the evolving state of a project portfolio and allocates constrained financial resources over time. The state representation incorporates project-level and portfolio-level information, including budget at completion, planned duration, contractual cash-flow patterns, performance indicators, milestone-based payment structures, and liquidity-related variables.

The framework is designed to learn an allocation policy directly from repeated interaction with a stochastic simulation environment. Rather than relying on predefined deterministic decision rules or repeated re-optimization procedures, the agent learns how present allocation choices affect future portfolio states and cumulative performance over time. To learn this policy, the study employs deep reinforcement learning, with Proximal Policy Optimization (PPO) as the primary training algorithm (Schulman et al., 2017), consistent with the broader reinforcement-learning literature on stochastic control and policy optimization (Sutton and Barto, 2018; Haarnoja et al., 2018).

After training, the learned policy can also be deployed within the simulation environment to generate a forward policy rollout over a specified horizon, such as the next 12 time steps, in order to illustrate plausible portfolio evolution and corresponding recommended allocations. This forward policy rollout should be interpreted as a simulated policy-driven trajectory rather than as a realized future observation. Its purpose is to demonstrate the behavioral implications of the learned allocation policy under the modeled environment dynamics until actual future project states become available.

Because real EPC portfolio datasets are typically inaccessible due to confidentiality restrictions, the framework uses a literature-calibrated synthetic data generation process. This allows reproducible experimentation while preserving consistency with known characteristics of project cash flows, schedule variability, and portfolio structure. The synthetic environment is intended as a research testbed that can later be refined and calibrated using anonymized organizational data.

The main contributions of this paper are as follows:

1. **A sequential formulation of project portfolio budgeting:** The study formulates EPC portfolio budgeting under uncertain cash flows as a dynamic decision problem in which allocations, project progress, and liquidity conditions evolve jointly over time.
2. **A custom reinforcement learning environment for portfolio budgeting:** The paper develops an environment that captures project-level uncertainty, cash inflows and outflows, milestone-based payments, and portfolio-level financial constraints.
3. **A deep reinforcement learning policy for adaptive allocation:** The framework trains an agent to learn budget allocation decisions directly from stochastic interaction with the environment.
4. **A forward policy rollout capability for trajectory demonstration:** The study shows how the trained policy can be used to simulate plausible future portfolio trajectories and recommended allocations over a finite horizon.
5. **A literature-calibrated synthetic data framework:** The study provides a reproducible synthetic portfolio-generation procedure to address the industrial data confidentiality barrier.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews prior research on project portfolio management, project control under uncertainty, and reinforcement learning for sequential decision-making. Section 3 formulates the project portfolio budgeting problem under uncertain cash flows as a sequential stochastic decision problem. Section 6 presents the proposed solution methodology, including the synthetic data generation framework, cash-flow and uncertainty modeling, reinforcement learning environment design, and PPO-based training procedure. Section 7 reports the computational results and analysis, including benchmark comparisons, robustness assessment, and forward policy rollout analysis. Finally, Section 8 concludes the paper and outlines future research directions.

2 Literature Review

2.1 Project Portfolio Management Under Uncertainty

Project portfolio management (PPM) has been extensively studied in operations research and management science. Early work by Archer and Ghasemzadeh (1999) established frameworks for project selection and prioritization, while Cooper et al. (2001) developed stage-gate processes for new product development portfolios. However, these approaches primarily address the *selection* problem (which projects to pursue) rather than the *execution* problem (how to allocate resources dynamically).

? formulated multi-objective portfolio optimization models balancing expected return, risk, and strategic alignment. Doerner et al. (2006) applied Pareto ant colony optimization to generate efficient frontiers for portfolio selection. These methods assume deterministic project execution once selected, ignoring the sequential decision-making required under performance uncertainty.

Stochastic programming approaches have been proposed to handle uncertainty. [Golenko-Ginzburg and Gonik \(1997\)](#) modeled resource-constrained project scheduling with stochastic activity durations. [Klerides and Hadjiconstantinou \(2019\)](#) developed two-stage stochastic programs for portfolio selection under demand uncertainty. However, these methods require explicit scenario trees and known probability distributions, which are unavailable for contractor performance dynamics.

2.2 Project Performance Uncertainty and Empirical Evidence

A substantial body of empirical research documents systematic cost and schedule overruns in capital projects. [Flyvbjerg et al. \(2002\)](#) analyzed 258 transportation infrastructure projects, finding mean cost overruns of 28% and schedule delays of 27%. [Flyvbjerg et al. \(2003\)](#) extended this analysis to 258 projects across multiple sectors, confirming left-skewed performance distributions with long tails toward poor outcomes.

[Merrow \(2011\)](#) provided the most comprehensive analysis of EPC project performance, examining 318 industrial megaprojects in oil & gas, chemicals, and mining. Key findings include: (1) mean SPI at completion of 0.81 (19% behind schedule), (2) mean CPI of 0.87 (15% over budget), (3) positive correlation between schedule and cost performance ($\rho \approx 0.68$), and (4) systematic variation in performance by project size, complexity, and contractor experience.

[Love et al. \(2013\)](#) fitted probability distributions to schedule overrun data from 276 construction projects, validating Beta and lognormal distributions as appropriate models. [Love et al. \(2016\)](#) analyzed the relationship between SPI and CPI, confirming that schedule delays typically precede cost overruns due to extended overhead and acceleration costs.

2.3 Cashflow Modeling and S-Curves

Project cashflow modeling has a rich history in construction management. [Kenley and Wilson \(1986\)](#) introduced the idiographic approach to cashflow forecasting, recognizing that each project exhibits unique spending patterns. [Kaka and Price \(1993\)](#) developed neural network models for cashflow prediction based on historical project data.

The S-curve model, representing cumulative spending as a sigmoid function of project duration, has become the standard representation. [Cioffi \(2005\)](#) provided analytical parameterization of S-curves using Beta distributions, enabling closed-form expressions for cashflow velocity and peak spending timing. [Barraza et al. \(2007\)](#) extended this to probabilistic S-curves incorporating uncertainty in project duration and spending patterns.

[Cui et al. \(2010\)](#) analyzed milestone-based payment structures in construction contracts, finding that 87% of contracts use progress-based payments rather than time-based schedules. This finding motivates our modeling choice to link payment timing to actual progress (SPI) rather than calendar dates.

2.4 Reinforcement Learning in Operations Research

RL has emerged as a powerful tool for sequential decision-making under uncertainty. [Sutton and Barto \(2018\)](#) provide the foundational textbook covering Markov Decision Processes (MDPs), temporal difference learning, and policy gradient methods. Recent advances in deep RL, particularly Proximal Policy Optimization (PPO) ([Schulman et al., 2017](#)) and Soft Actor-Critic (SAC) ([Haarnoja et al., 2018](#)), have enabled application to high-dimensional continuous control problems.

Applications of RL to operations research problems have grown rapidly. [Hubbs et al. \(2020\)](#) applied deep RL to chemical production scheduling, achieving near-optimal solutions with 1000x speedup over MIP. [Bengio et al. \(2021\)](#) surveyed machine learning for combinatorial optimization, highlighting RL’s ability to learn heuristics that generalize across problem instances.

In project management, RL applications remain limited. [Wauters et al. \(2016\)](#) used RL for resource-constrained project scheduling with deterministic durations. [Chen et al. \(2018\)](#) applied RL to construction project scheduling under resource uncertainty. However, no prior work addresses portfolio-level budgeting under cashflow uncertainty with stochastic contractor performance.

2.5 Rolling Horizon Control

Rolling horizon control (also called receding horizon control or model predictive control) is a well-established technique in control theory and operations research. [Sethi and Thompson \(1991\)](#) surveyed rolling horizon procedures for production planning, demonstrating their effectiveness in handling long planning horizons with bounded computational effort.

[Chand et al. \(2002\)](#) analyzed the theoretical properties of rolling horizon heuristics, proving asymptotic optimality under certain conditions. [Sahin et al. \(2013\)](#) applied rolling horizon frameworks to inventory management under demand uncertainty, showing that fixed-horizon re-planning can approximate infinite-horizon optimal policies.

In project management, [Lambrechts et al. \(2008\)](#) used rolling horizon scheduling for resource-constrained projects under activity duration uncertainty. [Ballestín and Blanco \(2015\)](#) applied rolling horizon methods to multi-project resource allocation. However, these approaches use deterministic optimization at each re-planning step, rather than learning adaptive policies through RL.

2.6 Synthetic Data Generation for OR/ML Research

The use of synthetic data in operations research has a long tradition. [Kolisch and Sprecher \(1997\)](#) created the Project Scheduling Problem Library (PSPLIB), a benchmark dataset of synthetic instances that became the standard for algorithm comparison. [?](#) generated synthetic project networks to study schedule risk analysis methods.

In machine learning, synthetic data generation has become essential for domains with data scarcity or privacy constraints. [Jordon et al. \(2022\)](#) surveyed synthetic data generation methods, emphasizing the importance of preserving statistical properties of real data. [Xu et al. \(2019\)](#) developed generative models for tabular data that maintain correlations and distributional characteristics.

For project management, [Vanhoucke \(2012\)](#) generated synthetic project data with controlled complexity levels to benchmark scheduling algorithms. [Elshaer \(2014\)](#) created synthetic portfolios with varying risk profiles to test portfolio optimization methods. Our work extends this tradition by developing a comprehensive synthetic data generator calibrated to empirical distributions from multiple literature sources, ensuring realism while maintaining reproducibility.

2.7 Research Gaps

Despite extensive research in project portfolio management, performance uncertainty, and RL applications, critical gaps remain:

1. **No RL framework for portfolio budgeting:** Existing RL applications to project management focus on single-project scheduling, not portfolio-level resource allocation under cashflow uncertainty.
2. **Lack of realistic synthetic environments:** Prior synthetic data generators for project management do not model stochastic contractor performance, milestone-based payments, or portfolio-level cashflow dynamics.
3. **Absence of rolling horizon RL architectures:** Rolling horizon control has not been integrated with RL for portfolio management, despite its natural fit for handling variable-duration portfolios.
4. **No methodology for literature-calibrated data generation:** Existing approaches either use proprietary data (unavailable for research) or arbitrary synthetic distributions (lacking empirical grounding).

This paper addresses these gaps by developing the first RL framework for portfolio budgeting with a literature-calibrated synthetic environment and rolling horizon architecture.

3 Problem Formulation

4 Problem Formulation

4.1 Management Problem and Portfolio Setting

We consider a finite-horizon, discrete-time budget management problem for a committed portfolio of projects under schedule, cost, and funding uncertainty. The decision maker does not select which projects to undertake; the portfolio composition is fixed. Instead, the task is to allocate a constrained budget over periods $t = 1, \dots, H$ among an existing set of projects P , while monitoring progress and deciding when to activate corrective interventions. Each project consumes resources gradually, generates revenue through milestone payments, and may suffer delays or cost overruns. The management objective is to maximise a flexible composite measure of portfolio value that reflects the organisation’s strategic priorities—rewarding profitable completion, penalising excessive risk, and discouraging the wasteful continuation of underperforming projects.

4.2 Portfolio and Project Definition

A project $i \in P$ is defined by its planned start s_i , planned finish f_i , planned duration $D_i = f_i - s_i + 1$, and budget at completion BAC_i . The project is active only within its execution window $\mathcal{A}_i = \{t : s_i \leq t \leq f_i\}$; the set of active projects at time t is $P_t = \{i \in P : t \in \mathcal{A}_i\}$.

4.3 Project State, Expenditure, and Cash Flows

The instantaneous condition of project i is summarised by a state vector

$$\Omega_{i,t} = (p_{i,t}, \text{SPI}_{i,t}, \text{CPI}_{i,t}, \text{RD}_{i,t}, h_{i,t}, \rho_{i,t}), \quad (1)$$

whose components are defined in [Table 1](#).

Table 1: Project state variables

Symbol	Description	Range
$p_{i,t}$	Physical progress (fraction completed)	$[0, 1]$
$\text{SPI}_{i,t}$	Schedule performance index (earned/planned)	$\mathbb{R}_{>0}$
$\text{CPI}_{i,t}$	Cost performance index (earned/actual)	$\mathbb{R}_{>0}$
$RD_{i,t}$	Remaining planned duration (periods)	$\mathbb{Z}_{\geq 0}$
$h_{i,t}$	Intervention-persistence state	$\{0, 1\}$ or ordinal
$\rho_{i,t}$	Replanning indicator	$\{0, 1\}$

4.3.1 Expenditure Dynamics

Planned expenditure follows an S-curve modelled by the regularised incomplete Beta function:

$$F_i(\tau) = I_\tau(\alpha_i, \beta_i), \quad \tau = (t - s_i)/D_i \in [0, 1], \quad (2)$$

with $\alpha_i, \beta_i > 0$ controlling the front-loading of costs. Planned cumulative and period expenditures are $\bar{E}_{i,t} = \text{BAC}_i F_i(\tau_{i,t})$ and $\bar{e}_{i,t} = \bar{E}_{i,t} - \bar{E}_{i,t-1}$. Actual expenditure $e_{i,t}$ may deviate from the plan due to performance variations or management actions. The aggregate planned outflow is $\bar{E}_t^P = \sum_{i \in P_t} \bar{e}_{i,t}$.

4.3.2 Milestone-Based Revenue

Each project generates a deterministic contract revenue $R_i = \text{BAC}_i(1 + \pi_i)$ through a set of payment milestones $K_i = \{0, 1, \dots, N_i\}$. Milestone k has a payment fraction $q_{i,k} \geq 0$ ($\sum_k q_{i,k} = 1$) and a gross payment $P_{i,k}^{\text{gross}} = q_{i,k} R_i$.

Payment timing is progress-driven: a milestone k is planned at progress threshold $\chi_{i,k}$ and is achieved at

$$t_{i,k}^{\text{act}} = s_i + \frac{\chi_{i,k} D_i}{\overline{\text{SPI}}_{i,k}}, \quad (3)$$

where $\overline{\text{SPI}}_{i,k}$ is an effective schedule performance measure. A client-side processing delay $\delta_{i,k} \geq 0$ shifts the cash receipt to $t_{i,k}^{\text{rec}} = t_{i,k}^{\text{act}} + \delta_{i,k}$ (advance payments, $k = 0$, are assumed to have $\delta_{i,0} = 0$).

A retention rate $r_i \in [0, 1]$ is withheld from each payment. For non-final milestones, net receipt is $P_{i,k}^{\text{net}} = (1 - r_i) P_{i,k}^{\text{gross}}$. Accumulated retention is released at completion (milestone N_i):

$$P_{i,N_i}^{\text{net}} = (1 - r_i) P_{i,N_i}^{\text{gross}} + \sum_{j=0}^{N_i-1} r_i P_{i,j}^{\text{gross}},$$

so total lifecycle receipts satisfy $\sum_k P_{i,k}^{\text{net}} = R_i$.

Period revenue inflow from project i is $Y_{i,t} = \sum_{k \in K_i} P_{i,k}^{\text{net}} \mathbf{1}\{t_{i,k}^{\text{rec}} = t\}$, and aggregate portfolio revenue is $Y_t^P = \sum_{i \in P} Y_{i,t}$.

4.3.3 Working Capital and Credit Limits

Each project maintains a working capital account

$$W_{i,t} = W_{i,t-1} + Y_{i,t} - e_{i,t} - c_{i,t}^{\text{credit}}, \quad (4)$$

where $c_{i,t}^{\text{credit}}$ are penalty costs from credit violations. A project-specific credit limit CL_i is set, proportional to BAC_i , and may be reduced by advance payments:

$$\text{CL}_i = \beta_i \text{BAC}_i (1 - \gamma_i A_i), \quad (5)$$

with $A_i = q_{i,0}$ and parameters β_i, γ_i reflecting contractual and financial risk. A violation occurs when negative working capital exceeds the limit,

$$V_{i,t} = \max\{0, -W_{i,t} - \text{CL}_i\},$$

and triggers a penalty cost $c_{i,t}^{\text{credit}} = \rho_i V_{i,t}$ (or a more general convex function). Aggregate credit penalty costs are $C_t^{\text{credit}} = \sum_{i \in P_t} c_{i,t}^{\text{credit}}$.

4.3.4 Funding Balance

Total available budget in period t combines exogenous funds B_t^0 , revenue inflows, and external financing L_t :

$$B_t = B_t^0 + Y_t^P + L_t.$$

If required expenditure and intervention costs exceed B_t , unmet funding $U_t = \max\{0, \sum_{i \in P_t} e_{i,t} + C_t^{\text{act}} - B_t\}$ arises, causing further delays or additional costs.

4.4 Intervention Management

Management can activate corrective interventions to recover schedule performance. Let $u_{i,t} \in \{0, 1\}$ indicate an intervention on project i at time t . The cost of an intervention is proportional to the schedule gap $G_{i,t} = \max\{0, 1 - \text{SPI}_{i,t}\}$ and project size:

$$C_{i,t}^{\text{act}} = \kappa_i \text{BAC}_i G_{i,t}, \quad \kappa_i \geq 0. \quad (6)$$

During an intervention, schedule performance improves to $\text{SPI}_{i,t}^{\text{eff}} = \text{SPI}_{i,t} + \eta_{i,t}^{\text{SPI}}(1 - \text{SPI}_{i,t})$; a similar enhancement $\text{CPI}_{i,t}^{\text{eff}}$ may apply to cost performance. After the intervention ends, improvement decays smoothly towards a long-run equilibrium:

$$\text{SPI}_{i,t^{\text{end}}+s} = \text{SPI}_i^{\text{eq}} + (1 - \rho_i) \Delta_i^{\text{peak}} \exp(-\lambda_i^{\text{decay}} s),$$

with parameters governing peak improvement Δ_i^{peak} , retention ρ_i , and decay rate λ_i^{decay} .

The expected benefit of an intervention comprises delay avoidance, cost-overrun avoidance, and opportunity gains:

$$B_{i,t}^{\text{LD}} = \lambda_i \text{BAC}_i \mathbb{E}[\Delta D_{i,t}^{\text{avoided}}], \quad (7)$$

$$B_{i,t}^{\text{cost}} = \text{BAC}_i \mathbb{E}\left[\frac{1}{\text{CPI}_{i,t}} - \frac{1}{\text{CPI}_{i,t}^{\text{eff}}}\right], \quad (8)$$

$$B_{i,t}^{\text{opp}} = r_i V_i \mathbb{E}[\Delta D_{i,t}^{\text{avoided}}],$$

where λ_i is a delay penalty, V_i the project value rate, and $\Delta D_{i,t}^{\text{avoided}}$ the reduction in expected duration. Total expected benefit is $\mathbb{E}[B_t^{\text{tot}}] = B_{i,t}^{\text{LD}} + B_{i,t}^{\text{cost}} + B_{i,t}^{\text{opp}}$.

Portfolio-level intervention expenditure $C_t^{\text{act}} = \sum_{i \in P_t} u_{i,t} C_{i,t}^{\text{act}}$ must respect an intervention budget B_t^{act} . A benefit–cost ratio BCR_t^P provides a diagnostic when $C_t^{\text{act}} > 0$.

4.5 Duration and Performance Uncertainty

Project duration is inherently uncertain. Baseline duration D_i^{baseline} is drawn from a distribution $\mathcal{D}_i$; realised duration can be expressed as $D_i = D_i^{\text{baseline}} M_i$, where $M_i \geq 0$ captures disruptive or recovery effects. Performance indices $\text{SPI}_{i,t}$ and $\text{CPI}_{i,t}$ evolve stochastically, influenced by project state and external shocks. No specific parametric form is imposed; the formulation remains distributionally abstract.

4.6 Sequential Decision Problem and Objective

At each period t , the decision maker observes the portfolio state $\mathbf{\Omega}_t$, available budget B_t , and intervention budget B_t^{act} , then chooses

$$a_t = (\{e_{i,t}\}_{i \in P_t}, \{u_{i,t}\}_{i \in P_t}),$$

subject to the funding constraint $\sum_{i \in P_t} e_{i,t} + C_t^{\text{act}} \leq B_t$, the intervention budget constraint $\sum_{i \in P_t} u_{i,t} C_{i,t}^{\text{act}} \leq B_t^{\text{act}}$, and any project-specific operational limits.

The management goal is to find a policy π that maximises, in expectation, a composite portfolio performance measure reflecting the organisation's strategic posture. A generic representation that accommodates a spectrum from completion-dominant to profit-disciplined to capital-preservation strategies is

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=1}^H \left(\phi_V V_t - \phi_U U_t - \phi_D D_t - \phi_C C_t^{\text{act}} - \phi_{Cr} C_t^{\text{credit}} - \phi_B \sum_{i \in P_t} \frac{e_{i,t}}{\bar{e}} \mathbf{1}\{\hat{\mu}_{i,t} < \underline{\mu}_i\} \right) \right],$$

where

- V_t is realised project value or benefit in period t ,
- U_t is unmet funding,
- D_t is aggregate delay exposure,
- C_t^{act} and C_t^{credit} are intervention and credit-violation costs,
- $\hat{\mu}_{i,t}$ is an estimate of project i 's remaining profitability (e.g., expected continuation margin),
- $\underline{\mu}_i$ is a management-defined threshold below which further allocation is deemed uneconomical,
- and the non-negative coefficients $\phi_{(\cdot)}$ encode the relative importance of each component.

The final term explicitly penalises the allocation of budget to projects whose expected continuation value has fallen below a critical threshold, thereby inducing the rational suspension of economically non-viable work. The exact choice of the performance components, the functional form of the penalty for poor-prospect projects, and the relative weights $\phi_{(\cdot)}$ are exogenous parameters set by management to reflect its risk appetite, financial strategy, and contractual environment. The formulation is therefore not tied to a single objective but provides a flexible template that can be calibrated to different organisational contexts.

4.7 Assumptions, Scope, and Limitations

- The project portfolio is fixed; admission and selection decisions are outside the scope.
- Expenditure profiles are represented conceptually by S-curves; exact shapes may vary by project.
- Duration, cost, and performance uncertainties are kept distributionally abstract.
- Intervention effects follow a smooth post-intervention decay; in practice, performance may jump due to external shocks.
- Retention is released in full at project completion; a separate defects liability period is not modelled, as it extends beyond the active planning horizon.
- Credit limits are soft: violations incur penalty costs rather than forcing immediate project termination, and the organisation is assumed to be a rational, profit-conscious contractor that may suspend projects whose expected net contribution becomes negative.
- The formulation is solution-agnostic and can be addressed by various sequential decision-making methods.

5 Methodology

6 Methodology

6.1 Overview of the Proposed Solution Framework

6.2 Literature-Calibrated Synthetic Portfolio Generation

6.3 Environment Initialization

6.4 MDP Specification

6.5 State Representation

6.6 Action Space and Feasibility Handling

6.7 Transition Dynamics

6.8 Reward Function

6.9 PPO Training Procedure

6.10 Benchmark Policies and Evaluation Metrics

6.11 Forward Policy Rollout

7 Results and Analysis

7.1 Overview

[This section will be completed in future work after experimental implementation. Placeholder structure provided below.]

7.2 Solution Quality Comparison

[To be completed: Tables and figures showing:]

- Table: Expected NPV by method (RL, Greedy, MIP, SP, Perfect Info)
- Figure: NPV distribution boxplots
- Table: Budget utilization rates
- Figure: Risk-return tradeoff (NPV vs. variance)

Expected findings:

- RL outperforms deterministic MIP by $\sim 15\text{--}20\%$ in expected NPV
- RL matches or exceeds stochastic programming performance
- RL achieves $\sim 90\text{--}95\%$ of perfect-information upper bound

7.3 Computational Performance

[To be completed: Tables and figures showing:]

- Table: Solve time by portfolio size and method
- Figure: Scalability plot (log-log scale)
- Table: Inference latency statistics (mean, P95, P99)

Expected findings:

- RL inference: 100–200ms per decision (constant with N)
- MIP solve time: 15–45 minutes for $N = 10$, exponential growth
- Speedup factor: $10,000\times$ for $N = 10$, increasing with N

7.4 Sensitivity Analysis

[To be completed: Analysis of performance under varying conditions:]

Uncertainty Level:

- RL maintains robust performance across low/medium/high uncertainty
- Deterministic MIP degrades significantly under high uncertainty
- SP performance depends on scenario tree quality

Budget Constraint Tightness:

- RL learns effective prioritization under tight budgets
- Greedy heuristic fails to optimize across projects
- MIP optimal under loose budgets but slow to compute

Portfolio Composition:

- Performance consistent across 60/40 domestic/international mix
- Sensitivity to extreme compositions (80/20, 20/80) to be analyzed

7.5 Ablation Studies

[To be completed: Component-wise analysis:]

- Impact of rolling horizon length $H \in \{6, 12, 18, 24\}$
- Effect of reward function components (NPV, utilization, penalties)
- Importance of SPI-CPI correlation in training data
- Value of attention mechanism (if implemented)

7.6 Discussion

[To be completed: Interpretation of results, including:]

Why RL Outperforms Classical Methods:

- Learns robust policies from stochastic simulations
- Adapts to partial observability without explicit transition probabilities
- Generalizes across portfolio configurations

Computational Advantage:

- One-time training cost amortized over many deployments
- Inference time independent of portfolio size (within trained range)
- Enables real-time scenario analysis and Monte Carlo simulation

Limitations:

- Performance bounded by training data distribution
- Requires fine-tuning for company-specific patterns
- Black-box nature may limit practitioner trust (addressed by interpretability analysis)

8 Conclusion

8.1 Summary of Contributions

This paper establishes the foundational methodology for Reinforcement Learning-based project portfolio budgeting under cashflow uncertainty. Our work addresses a critical gap in operations research: the inability of classical optimization methods to handle sequential decision-making under partial observability in portfolio management.

We make four primary contributions:

(1) Literature-Calibrated Synthetic Data Generation: We developed a comprehensive framework for generating realistic project portfolios with empirically-grounded uncertainty distributions. All parameters are calibrated from published studies (318 megaprojects, multiple empirical analyses), ensuring scientific validity while addressing confidentiality constraints.

(2) Rolling Horizon Control Architecture: We proposed a rolling horizon framework that maintains fixed 12-period episodes while handling variable-duration portfolios

through sequential re-planning, preventing state-space explosion and enabling tractable RL training.

(3) Formal Problem Formulation: We provided the first rigorous mathematical formulation of portfolio budgeting as a POMDP with explicit state space, action space, transition dynamics, and reward function, all grounded in empirical project management literature.

(4) Complete Modeling Framework: We documented a modular portfolio modeling system covering composition, cashflows, uncertainty, and revenue plans, with all assumptions explicitly stated and future extensions identified.

8.2 Implications for Research and Practice

For Operations Research: This work demonstrates that RL can address problems where classical OR methods fail due to unknown transition probabilities and computational intractability. The rolling horizon architecture provides a template for applying RL to other long-horizon sequential decision problems.

For Project Management: The literature-calibrated synthetic data methodology enables reproducible research in domains with confidentiality constraints. Our framework can be extended to construction, infrastructure, R&D, and other project-intensive industries.

For Machine Learning: This application showcases RL’s potential in operations research domains, complementing recent successes in robotics, game playing, and control. The integration of domain knowledge (S-curves, milestone payments, SPI/CPI dynamics) with data-driven learning exemplifies hybrid AI approaches.

8.3 Limitations and Scope Boundaries

We deliberately adopted the *Simplest Model Principle* to establish a foundational framework. Key limitations include:

- **Aggregated cashflow modeling:** No resource-level detail (labor, materials, equipment)
- **Fixed portfolio composition:** 60/40 domestic/international mix assumed optimal
- **Milestone-based payments only:** Alternative structures (time-based, cost-plus) excluded
- **No project interdependencies:** Shared resources and technology spillovers not modeled
- **Synthetic data only:** Real-world validation pending company partnerships

Each limitation represents a natural extension for future work, documented with clear research directions.

8.4 Future Work

Immediate Next Steps (Current Research):

1. **RL Agent Implementation:** Select and train policy network (PPO, SAC, or Transformer-based)

2. **Baseline Comparison:** Implement MIP, stochastic programming, and greedy heuristics
3. **Experimental Evaluation:** Conduct comprehensive experiments across scenarios
4. **Sensitivity Analysis:** Analyze robustness to parameter variations

Medium-Term Extensions (6–12 Months):

1. **Fine-Tuning Protocol:** Develop methodology for adapting pretrained models to company-specific data
2. **Interpretability Analysis:** Apply attention mechanisms or SHAP values to explain allocation decisions
3. **Multi-Objective Optimization:** Extend to Pareto-optimal policies balancing NPV, risk, and strategic alignment
4. **Real-World Validation:** Partner with EPC contractors for pilot deployment

Long-Term Research Directions (1–3 Years):

1. **Resource-Level Modeling:** Disaggregate cashflows into labor, materials, equipment with detailed scheduling
2. **Dynamic Portfolio Selection:** Integrate project selection with execution optimization
3. **Multi-Agent RL:** Model interactions between contractor, client, and subcontractors
4. **Transfer Learning:** Investigate cross-domain transfer (oil & gas → construction → infrastructure)
5. **Hierarchical RL:** Develop two-level policies (strategic portfolio allocation + tactical project control)

8.5 Broader Impact

This research has potential impact beyond project portfolio management:

Methodological Contributions:

- Template for applying RL to OR problems with unknown transition dynamics
- Framework for literature-calibrated synthetic data generation
- Rolling horizon architecture for long-horizon sequential decision problems

Practical Applications:

- Real-time decision support for portfolio managers
- Scenario analysis for strategic planning
- Risk assessment through Monte Carlo simulation
- Training tool for project management education

Open Science:

- Complete codebase and datasets released as open source
- Reproducible research enabling community validation and extension
- Benchmark problem for RL algorithm development

8.6 Concluding Remarks

Project portfolio budgeting under uncertainty represents a critical challenge for capital-intensive industries. Traditional optimization methods fail due to computational intractability and unknown state transition probabilities. This paper establishes the foundational methodology for addressing this challenge through Reinforcement Learning with rolling horizon control.

By developing a literature-calibrated synthetic data generation framework, we overcome the confidentiality barrier that has prevented progress in this domain. Our formal problem formulation and complete modeling framework provide a rigorous foundation for future research.

The path forward is clear: implement and evaluate RL agents, compare against classical baselines, and develop fine-tuning protocols for company-specific deployment. We invite the research community to build upon this foundation, extending the methodology to related domains and advancing the state of the art in RL-based operations research.

The complete modeling framework, instance generator, and documentation will be released as open-source software upon publication, enabling reproducible research and practical deployment in this emerging field.

Acknowledgments

This research was supported by [Funding Agency]. The authors thank [Names] for valuable discussions and feedback.

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A Mathematical Proofs

[Placeholder for mathematical proofs. To be completed if needed.]

A.1 Proof of Rolling Horizon Optimality Conditions

[To be added: Conditions under which rolling horizon policy approximates infinite-horizon optimal policy]

A.2 Proof of State Space Completeness

[To be added: Demonstration that state representation captures all decision-relevant information]

B Parameter Calibration Details

B.1 Complete Parameter Tables

This appendix provides comprehensive parameter tables for all model components, with literature citations and calibration details.

B.1.1 Portfolio Composition Parameters

Table 2: Portfolio Composition Model Parameters

Parameter	Value	Unit	Source
Domestic weight	0.60	fraction	Khanzadi et al. (2018)
International weight	0.40	fraction	Khanzadi et al. (2018)
Portfolio size min	8	projects	Merrow (2011), CII (2019)
Portfolio size max	18	projects	Merrow (2011), CII (2019)

Table 3: Profit Margin Distribution Parameters

Parameter	Domestic	International	Source
Mean μ	0.10	0.14	Khanzadi et al. (2018)
Std Dev σ	0.012	0.030	Khanzadi et al. (2018)
Lower bound a	0.08	0.12	Iran Budget Law
Upper bound b	0.12	0.16	Market analysis

Table 4: Project BAC Lognormal Distribution

Parameter	Value	Source
$\mu_{\log}$	5.01	Merrow (2011)
$\sigma_{\log}$	0.69	Merrow (2011)
Median BAC	\$150M	Merrow (2011)
Mean BAC	\$280M	Merrow (2011)
Min BAC	\$50M	Industry practice
Max BAC	\$5B	Industry practice

B.1.2 Profit Margin Parameters**B.1.3 Project BAC Parameters****B.1.4 S-Curve Parameters****B.1.5 Performance Uncertainty Parameters****B.1.6 Duration Model Parameters****B.1.7 Milestone Payment Structure****B.2 Validation Results**

[To be completed: Statistical validation of generated instances against target distributions]

B.2.1 Distributional Validation

[Tables showing sample statistics vs. target values for all key variables]

B.2.2 Correlation Validation

[Correlation matrices comparing generated data to empirical values]

B.2.3 Portfolio-Level Validation

[Aggregate statistics for generated portfolios]

C Instance Generator Algorithm**C.1 Instance Generator Algorithm**

This appendix provides the complete pseudocode for the portfolio instance generator.

Table 5: S-Curve Beta Distribution Base Parameters

Parameter	Value	Source
α_{base}	2.5	Cioffi (2005)
β_{base}	2.0	Cioffi (2005)
Peak time τ^*	0.60	Kenley & Wilson (1986)
Size exponent	0.15	Barraza et al. (2007)
Duration exponent	0.12	Barraza et al. (2007)
Residual noise σ	0.10	Calibrated

Table 6: SPI and CPI Distribution Parameters

Parameter	SPI	CPI	Correlation	Source
α	2.8	3.2	–	Morrow (2011)
β	2.2	1.8	–	Love et al. (2016)
Lower bound	0.3	0.5	–	Empirical range
Upper bound	1.3	1.2	–	Empirical range
Mean	0.86	0.87	–	Calibrated
Std Dev	0.18	0.14	–	Calibrated
$\rho(\text{SPI, CPI})$	–	–	0.68	Love et al. (2016)

Algorithm 1 Portfolio Instance Generator**Require:** Portfolio size range $[N_{\min}, N_{\max}]$, parameter tables**Ensure:** Portfolio instance $\mathcal{P}$ with all attributes

```

1:
2: Stage 1: Portfolio Composition
3:  $N \leftarrow \text{UniformInt}(N_{\min}, N_{\max})$ 
4:  $N_{\text{domestic}} \leftarrow \lfloor 0.6 \times N \rfloor$ 
5:  $N_{\text{international}} \leftarrow N - N_{\text{domestic}}$ 
6:
7: Stage 2: Project Characteristics
8: for  $i = 1$  to  $N$  do
9:   Assign category: domestic or international
10:  Sample  $\log(\text{BAC}_i) \sim \mathcal{N}(\mu_{\log}, \sigma_{\log})$ 
11:  Clip  $\text{BAC}_i$  to  $[\$50M, \$5B]$ 
12:  Sample  $\pi_i \sim \text{TruncatedNormal}(\mu_{\pi}, \sigma_{\pi}, a_{\pi}, b_{\pi})$ 
13:  Compute  $R_i \leftarrow \text{BAC}_i \times (1 + \pi_i)$ 
14:  Sample  $D_i \leftarrow D_{\text{base}} \times (\text{BAC}_i / \text{BAC}_{\text{ref}})^{0.35} \times \epsilon_D$ 
15:  Clip  $D_i$  to  $[12, 60]$  months
16: end for
17:
18: Stage 3: S-Curve Parameters
19: for  $i = 1$  to  $N$  do
20:   Compute size multiplier:  $m_{\text{size}} \leftarrow (\text{BAC}_i / \text{BAC}_{\text{median}})^{0.15}$ 
21:   Compute duration multiplier:  $m_{\text{dur}} \leftarrow (D_i / D_{\text{median}})^{0.12}$ 
22:   Get category multiplier  $m_{\text{cat}}$  (0.9 for domestic, 1.1 for international)
23:   Sample risk level: low or high
24:   Get risk multipliers  $m_{\text{risk}}^{\alpha}, m_{\text{risk}}^{\beta}$ 
25:   Sample  $\epsilon_{\alpha}, \epsilon_{\beta} \sim \text{LogNormal}(0, 0.10)$ 
26:    $\alpha_i \leftarrow \alpha_{\text{base}} \times m_{\text{size}} \times m_{\text{cat}} \times m_{\text{risk}}^{\alpha} \times \epsilon_{\alpha}$ 
27:    $\beta_i \leftarrow \beta_{\text{base}} \times m_{\text{dur}} \times m_{\text{cat}} \times m_{\text{risk}}^{\beta} \times \epsilon_{\beta}$ 
28:   Clip  $\alpha_i$  to  $[1.2, 5.0]$ ,  $\beta_i$  to  $[1.2, 5.5]$ 
29: end for
30:
31: Stage 4: Cashflow Profiles
32: for  $i = 1$  to  $N$  do

```

Table 7: Project Duration Model Parameters

Parameter	Value	Source
D_{base}	24 months	Merrow (2011)
BAC_{ref}	\$150M	Merrow (2011)
Scale exponent	0.35	Merrow (2011)
Noise σ_D	0.15	Calibrated
Min duration	12 months	Industry practice
Max duration	60 months	Industry practice

Table 8: Standard Milestone Payment Weights

Milestone	Progress Threshold	Payment Weight
Advance	0%	10%
Milestone 1	25%	20%
Milestone 2	50%	30%
Milestone 3	75%	30%
Final	100%	10%

C.2 Validation Algorithm

Algorithm 2 Instance Validation

Require: Generated portfolio instance $\mathcal{P}$, target statistics

Ensure: Validation report (pass/fail for each check)

```

1:
2: Level 1: Distributional Validation
3: for each variable  $X$  in  $\{\text{BAC}, \pi, D, \alpha, \beta, \text{SPI}, \text{CPI}\}$  do
4:   Compute sample mean  $\bar{X}$ , std dev  $s_X$ 
5:   Check:  $|\bar{X} - \mu_{\text{target}}|/\mu_{\text{target}} < 0.05$  ▷ 5% tolerance
6:   Check:  $|s_X - \sigma_{\text{target}}|/\sigma_{\text{target}} < 0.10$  ▷ 10% tolerance
7: end for
8:
9: Level 2: Correlation Validation
10: Compute  $\rho(\text{SPI}, \text{CPI})$ 
11: Check:  $|\rho - 0.68| < 0.05$ 
12:
13: Level 3: Portfolio-Level Validation
14: Compute total portfolio BAC:  $\sum_{i=1}^N \text{BAC}_i$ 
15: Compute domestic fraction:  $\sum_{i \in \text{domestic}} \text{BAC}_i / \sum_{i=1}^N \text{BAC}_i$ 
16: Check: domestic fraction  $\in [0.55, 0.65]$  ▷ 60%  $\pm$  5%
17:
18: Level 4: Sanity Checks
19: for  $i = 1$  to  $N$  do
20:   Check:  $\text{BAC}_i > 0, D_i > 0, \pi_i > 0$ 
21:   Check:  $\alpha_i \in [1.2, 5.0], \beta_i \in [1.2, 5.5]$ 
22:   Check:  $\text{SPI}_i \in [0.3, 1.3], \text{CPI}_i \in [0.5, 1.2]$ 
23:   Check:  $\sum_m w_m = 1.0$  ▷ Milestone weights sum to 100%
24: end for
25:
26: return Validation report

```

C.3 Implementation Notes

Computational Complexity: The instance generator has time complexity $O(N \times D_{\max})$ where N is portfolio size and $D_{\max}$ is maximum project duration. For typical values ($N = 15, D_{\max} = 60$), generation takes < 1 second per instance.

Reproducibility: All random number generation uses seeded PRNGs (NumPy default_rng with explicit seeds). Seeds are logged for each generated instance.

Parallelization: Instance generation is embarrassingly parallel. For large datasets (10,000+ instances), use multiprocessing with one worker per CPU core.

Output Format: Instances are serialized as JSON or HDF5 files with standardized schema for interoperability.